Analysis and Design for Disaster Supply Provision System Based on the Transition of Required Information in the Noto Peninsula Earthquake*

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Abstract

Large scale earthquakes frequently cause serious damage in Japan, and past experiences have shown that the information and relief supplies required by disaster victims change over time. Recent advances in information and communication technology have improved information sharing during disasters, but have also created challenges such as the overloads or mismatches between actual needs and supplied goods. Therefore, this paper analyzes the transition of required information during the Noto Peninsula Earthquake using access data on the official website of Nanao City. The results show a clear shift from immediate survival related information, including water, food, and relief supplies, to recovery related information such as donations and support systems in later phases. Based on these findings, issues in disaster relief supply provision under modern communication environments are discussed, and the paper also proposes the key requirements for a Disaster Relief Supply Provision System that considers time dependent needs and supports efficient supply management.

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1 Introduction

Human society has experienced many large earthquakes, and these events have led to the accumulation of disaster-related records and experiences. Such knowledge has been shared over time and has supported the development of research activities and practical countermeasures. These efforts have contributed to improving disaster preparedness and recovery.

Especially in recent years, research in information and communication technology has begun to focus on the effective use of disaster records and memories. Advances in digital technology have made it possible to store, analyze, and share information more efficiently. As a result, past disaster experiences can now be reflected more directly in disaster response and management.

In Japan, the Great Hanshin Awaji Earthquake in 1995 occurred during the early period of Internet use. This event highlighted both the possibilities of Web system such as Email and the limitations of land phone systems in disaster situations. After that, Japan experienced several major earthquakes, including the Niigata Chuetsu Earthquake in 1994, the Great East Japan Earthquake in 2011, the Kumamoto Earthquake in 2016, and the Noto Peninsula Earthquake in 2024. Through these disasters, various problems in the developments of information and communication systems during emergencies became clear.

One of important research topics has been disaster information systems that consider changes in required information and supplies after a disaster. The information needed by affected people changes over time, especially in the early stage just after a disaster. Based on experiences from the Great East Japan Earthquake, one of the studies has examined methods to provide essential information quickly and continuously using heterogeneous communication networks, even when part of the infrastructure is damaged. (Shiratori, Uchida, Shibata, & Izumi, 2013)

Considered with these previous studies, this paper focuses on the transition of information required by disaster victims during the Noto Peninsula Earthquake that occurred in 2023. Then, the comparisons with other major earthquakes in Japan are also discussed to identify common features and differences for the future information and communication technology. Finally, this paper discusses the approaches for supporting the provision of essential supplies in the early post-disaster phase and explores approaches for the early recovery and development of supply chains.

In the followings, the major large-scale disasters in Japan and the features of Information Technology (IT) during these disasters are discussed in section II. Section III explains the previous studies of the required information in disasters, and section IV explains the required information in Noto Peninsula Earthquake using the records of Noto City HP accesses. Finally, Section V discusses the designs of the Disaster Supply Provision System with the transitions of the required information under the Noto Peninsula Earthquake, and Section VI presents the conclusions and future works in this research.

2 Major Large-Scale Disasters and Information Technology

Table 1 summarizes the major large-scale earthquakes that have occurred in Japan in recent years, including the number of fatalities, the magnitude, and the seismic intensity observed in each event. (International Seismological Centre, 2015) (United States Geological Survey, N.A.)

Earthquake	Date	Deaths	Missing	Magnitude	*JMA Seismic Intensity*
Great Hanshin Earthquake (Kobe)	January 17, 1995	6,434	N/A	6.9	Shindo 7
Niigata Chuetsu Earthquake	October 23, 2004	68	N/A	6.8	Shindo 7
Great East Japan Earthquake (Tohoku)	March 11, 2011	19,759	2,553	9.0 - 9.1	Shindo 7
Kumamoto Earthquake	April 16, 2016	277	N/A	7.0	Shindo 7
2024 Noto Peninsula Earthquake	January 1, 2024	703	2	7.5	Shindo 7

Table 1: Major Large-scale Earthquakes in Japan

These earthquakes clearly show that different types of damage have caused significant human losses. In the Great Hanshin Awaji Earthquake, a large number of deaths were mainly caused by the collapse of residential buildings, resulting in 6,434 fatalities. In contrast, the Great East Japan Earthquake caused catastrophic damage due to an unprecedented tsunami, leading to 19,759 fatalities, which remains one of the most severe disaster outcomes in recent history.

In more recent years, earthquakes such as the Kumamoto Earthquake and the Noto Peninsula Earthquake have also resulted in many deaths, primarily due to the collapse of houses and buildings. These events indicate that even with technological and social progress, structural damage continues to be a major factor contributing to human loss during earthquakes in Japan.

	Developments of IT	Subjects after Disaster
Great Hanshin Earthquake (Kobe)	Early period of Internet	Expectations of Internet (Email) during disasters/Loss of land phone network
Niigata Chuetsu Earthquake	Early period of cellular ph one	Expectations of cellular phone during disasters/Loss of land phone network
Great East Japan Earthquake (Tohoku)	Spread Internet and Cellula r Phone	Ensuring redundancy
Kumamoto Earthquake	Spread of SNS media	Excessive relief supplies
2024 Noto Peninsula Earthquake	LEO communication	Restriction of individual relief supplies

Table 2: Developments of IT and Major Subject after Disaster

From the perspective of information technology (IT), Japan has played a central role in advancing disaster-related research based on the accumulated memories and records of these events as shown in Table 2. The Great Hanshin Awaji Earthquake in 1995 occurred during the early period of Internet expansion. At that time, technologies such as web pages and email systems demonstrated the potential to function effectively during disasters. A valuable record described in Reference (Tanaka, 1996) reports that at Kobe University, information exchange was conducted using email while telephone networks and transportation systems were paralyzed. Also, while land phones could not available after the Niigata Chuetsu Earthquake in 2004, cellular phones became widely known as a useful tool during earthquakes in Japan. These experiences are considered to have triggered subsequent research on Internet-based disaster information systems, including systems that employed long-distance wireless

LAN technologies during the eruption of Mount Iwate. (Uchida, Asahi, & Shibata, Disaster Information System and Its Wireless Recovery Protocol, 2004)

The Great East Japan Earthquake in 2011 occurred after cellular phones and the Internet had become widespread. Although these technologies were widely available at the time, the disaster revealed their vulnerability under extreme conditions (Araemishi, Iwate Broadcasting Co.,Ltd., 2012), which led to increased recognition of the importance of redundancy in information and communication systems. At the same time, the availability of mobile devices also enabled the collection of a large amount of video and photographic data. As a result, systems for confirming personal safety and the use of SNS platforms began to be considered as practical disaster response measures.

During the Kumamoto Earthquake in 2016, communication methods were maintained to some extent due to previous technological developments. However, new challenges became evident, including the spread of misinformation and the excessive or uneven provision of relief supplies. (2016 Kumamoto Earthquake: One month record, 2016) These issues highlighted the importance of distributing accurate information and appropriate supplies according to actual needs in disaster-affected areas.

Furthermore, in the Noto Peninsula Earthquake of 2024, LEO (Law Earth Orbit) satellite communication services such as Starlink were introduced and made a significant contribution to the recovery of communication functions. While these technologies enabled rapid information sharing, the large volume of information from affected areas also created challenges. In some cases, individual supply support was restricted, and information provision became concentrated at local governments. (Ishikawa, 2024) These experiences suggest that issues related to information management and relief supply coordination immediately after a disaster remain important topics for continued examination.

3 Previous Studies

In Japan, it is widely recognized that the type of information required by disaster victims changes over time after a disaster. Based on this understanding, many local governments have developed disaster response manuals that consider the transition of required information from the immediate aftermath to later recovery phases. This recognition has been strongly influenced by research studies that analyzed actual disaster situations and the behavior of affected communities.

In fact, some representative studies reported the approaches with the transitions of required information based on experiences from the Great East Japan Earthquake. For example, the study (Uchida, Kazuo, & Shibata, Disaster Information System from Communication Traffic Analysis and Connectivity (Quick Report from Japan Earthquake and Tsunami on March 11th, 2011), 2011) investigated the transition of network conditions and information needs at Iwate Prefectural University, which functioned as an evacuation shelter after the disaster. This study examined how available communication resources changed over time and what kinds of information were required by disaster victims on a daily basis. It also reported practical efforts to provide necessary information through bulletin boards, reflecting the actual demands observed at each stage of the disaster response.

Another study (Shiratori, Uchida, Shibata, & Izumi, 2013) organized the transition of required information after disasters into a structured framework. The study proposed the seven disaster response stages as shown in Figure 1.

	Condition	Activity	Period
t_1	Normal		
t_2	Indication of Disaster	Indication, rumor etc	2 weeks before – Disaster
t_x	Occurrence of Disaster		During Disaster
t_3	Just after Disaster	Rescue, evacuation, safety information	Disaster – 2 days
t_4	Calm down from Disaster	Relief materials, safety information	3 days – 2 weeks
t_5	Restoration form Disaster	Restore lifeline, residences, and so on	3 weeks - several months
t_6	Revival		

Table 3: Proposed Disaster Response Stages in Great East Japan Earthquake

In Table 3, after the disaster happened at the stage t_x, the information such as rescue, evacuation, or safety information is necessary during the stage t_3. Therefore, the network connectivity is the most important, and the text contents are mainly used during the stage. Next, in the stage t_4, the information such as relief supplies is getting more important, and so throughput such VoIP (Voice over IP) is getting more important. Then, Figure 2 shows the required information through the time.

Subject	Necessary Information / time	t_1	t_2	t_x	t_3	t_4	t_5	t_6
Victim (Inside of the area)	Disaster Prevention	△	○					
	Evacuation Shelter		○		◎			
	Safety				◎	◎	○	△
	Stricken Area				◎	○		
	Traffic				○	◎	◎	
	Relief Supplies					◎	○	
	Public Service					○	◎	
	Lifeline					○	◎	
	Local Government					○	◎	
Relatives, Volunteer (Outside of the area)	Safety				◎	◎	○	
	Stricken Area				◎	○	△	
	Relief Supplies					○	◎	

Table 4: Proposed Disaster Response Stages in Great East Japan Earthquake

In Table 4, the forecast and evacuation information are required during the stage t_1, and evacuation information is not so much important. However, if there are some kinds of indications such as news or rumor about disaster during t_2, evacuation and disaster prevention becomes more important than usual. Then, if disaster occurs, very significant activities related of human lives such as rescue, evacuation, and safety status information come to be required. Since the information is much related to human lives during the stage of t_3, Disaster Information Network should be focused on the network connectivity more than throughput or latency. Also, text contents are mainly used for the disaster evacuation, resident safety and disaster status information.

Then, this study [4] focused on heterogeneous and multilayer communication network environments and proposed a Quality of Service (QoS) control method designed to maintain network

operation under disaster conditions. The proposed method aimed to ensure persistent network survival and early recovery of performance, while also considering changes in the priority of required information over time. This approach emphasized the importance of adaptive control mechanisms that respond to dynamically changing information demands.

However, the communication environment has continued to evolve since these earlier studies. The introduction of new communication technologies, such as satellite communication systems, and the widespread use of social networking services have significantly changed the way information is generated and shared during disasters. During the Noto Peninsula Earthquake, these changes led to new challenges, including information overload and the provision of support supplies that did not match the actual needs of disaster victims. In some cases, excessive or unsuitable relief supplies were delivered, creating additional burdens for affected areas.

These experiences indicate that it is no longer sufficient to manage only the transition of the required information in a disaster. The contents of support activities and the distribution of relief supplies also need to be managed appropriately. Efficient coordination based on accurate and update information remains an important issue for supporting effective recovery in the post disaster phase.

4 The Transition of Required Information in the Noto Peninsula Earthquake

Table 5 summarizes the top accessed keywords and other contents that showed a rapid increase in access on the official website of Nanao City, which is one of the affected areas, during the first four months after the earthquake. The figure reflects how information needs changed immediately after the disaster based on actual access logs.

Emergency Period; Required Information for Daily Necessities

Jan. 2024	Feb. 2024	Mar. 2024	Apr. 2024
3.5 (Million HP accesses)	2.0	1.2	0.7
Top Keywords: - Water and sewerage - Relief supplies - Disaster victim certificate - Housing support	Top Keywords: - Water and sewerage - Relief supplies - Disaster victim certificate - Government-funded demolition program - Housing support - Disaster relief donation/fund	Top Keywords: - Water and sewerage - Relief supplies - Disaster relief donation/fund - temporary disaster waste storage site - List of disaster support programs	Top Keywords: - Water and sewerage - Disaster relief donation/fund - Relief supplies - List of disaster support programs - temporary disaster waste storage site
Others; - Evacuation shelter - Account for donations - Bathing support - How to dispose of garbage - Risk assessment	Others; - Emergency housing - Bathing support - Public relations (PR) Nanao - How to dispose of garbage - Emergency support toilet -Self-funded demolition with later reimbursement	Others; - Government-funded demolition program - Disaster victim certificate - Public relations (PR) Nanao - Emergency housing - Self-funded demolition with later reimbursement	Others; - Disaster victim certificate - Government-funded demolition program - Public relations (PR) Nanao - Emergency housing - Recovery of septic tank

Table 5: Nanao City HP Accesses during Jan. 2024 to April. 2024

Before the earthquake, the website recorded approximately 300,000 accesses. After the earthquake, the number of accesses sharply increased to approximately 3,500,000, indicating that the municipal website became an important source of information for residents. During this period, the city council

of Nanao City reported that it was an emergency response phase, in which daily necessities were of primary importance. In January 2024, access was highly concentrated on information related to water supply and drainage, as well as relief supplies. In particular, information on food and drinking water, which are essential for daily life, was strongly demanded during the early post disaster period.

Table 6 presents the top accessed keywords and other contents on the Nanao City website from five to eight months after the earthquake.

Recovery Period; Required Information for Rebuilding Lives and Communities Programs

May. 2024	Jun. 2024	Jul. 2024	Aug. 2024
0.43 (Million HP accesses)	0.37	0.37	0.33
Top Keywords: - Disaster relief donation/fund - temporary disaster waste storage site - List of disaster support programs - Paypay (QR code cashless payment) - Government-funded demolition program	Top Keywords: - Disaster relief donation/fund - Accommodation facilities - temporary disaster waste storage site - List of disaster support programs - Paypay (QR code cashless payment) - Open meeting - Government-funded demolition program - Disaster victim certificate	Top Keywords: - Disaster relief donation/fund - List of disaster support programs - temporary disaster waste storage site - Government-funded demolition program - Accommodation facilities - Issaki Hoto Festival	Top Keywords: - Disaster relief donation from Taiwan - List of disaster support programs - Government-funded demolition program - Emergency housing - Issaki Hoto Festival - Bid information
Others; - Staff recruitment examination - Water and sewerage - Accommodation facilities - Emergency housing - Bid information	Others; - Water and sewerage - Emergency housing - Public relations (PR) Nanao - Bid information	Others; - Disaster victim certificate - City tax reductiion - Emergency housing - Bid information - Self-funded demolition with later reimbursement	Others; - Accommodation facilities - temporary disaster waste storage site - Recovery of disaster-damaged residential area - Free carry-in of disaster garbage - Nanao Port Festival - Subsidy for seismic retrofitting for damaged housing - Historic site of Nanao castle

Table 6: Nanao City HP Accesses during May. 2024 to Aug. 2024

Compared with the immediate post disaster phase, the total number of accesses decreased significantly from 3,500,000 to approximately 430,000. At the same time, a clear transition in information needs can be observed. The keywords with the highest access during this period were related to donations, disaster waste management, and public support programs for recovery. These results indicate that the focus of information demand shifted from emergency support for daily life to measures required for longer term recovery and reconstruction.

In addition, Nanao City reported an increase in access referrals from social networking services (SNS) such as LINE. Notably, content related to donation support from Taiwan showed a particularly high level of access from SNS accesses. This trend reflects recent changes in the information environment and disaster response behavior, which were not observed during the Great East Japan Earthquake in 2011. These results suggest that SNS platforms have become an important channel for information dissemination and access in recent Japanese disasters, further influencing the transition of required information over time.

5 Examples of Disaster Relief Supply Provision System

Unlike the Great East Japan Earthquake, communication networks remained functional during the Noto Peninsula Earthquake and the Kumamoto Earthquake. As a result, television reports and SNS rapidly transmitted information from disaster affected areas. While this enabled faster awareness, it also caused specific relief items to be delivered excessively by individuals. In some cases, large quantities of supplies such as diapers flooded the affected areas. This situation not only created storage and management problems but also placed additional pressure on administrative disaster relief activities that were responsible for distributing supplies to victims. Consequently, the restrictions on individual relief supply deliveries were introduced in certain regions, as reported in Reference (Ishikawa, 2024).

Based on these experiences, several problems must be addressed when designing future Disaster Relief Supply Provision Systems as the follows.

Subjects for Disaster Supply Provision System;
(1) Mismatch between the supplies required by victims and the relief supplies provided.
(2) Transition of required supplies through time
(3) Inconsistency in item labeling (such as bread, baguette, bagel, wheat, croissant, and so on)

To address these issues, future Disaster Relief Supply Provision Systems are expected to include several key functions as the follows.

Required functions for Disaster Supply Provision System;
(1) Triage considering real-time required supplies by victim's requests, media influences (HP, SNS, TV, and so on), and storage
(2) Real-time priority functions by the above triage functions
(3) Thesaurus-based categorization function

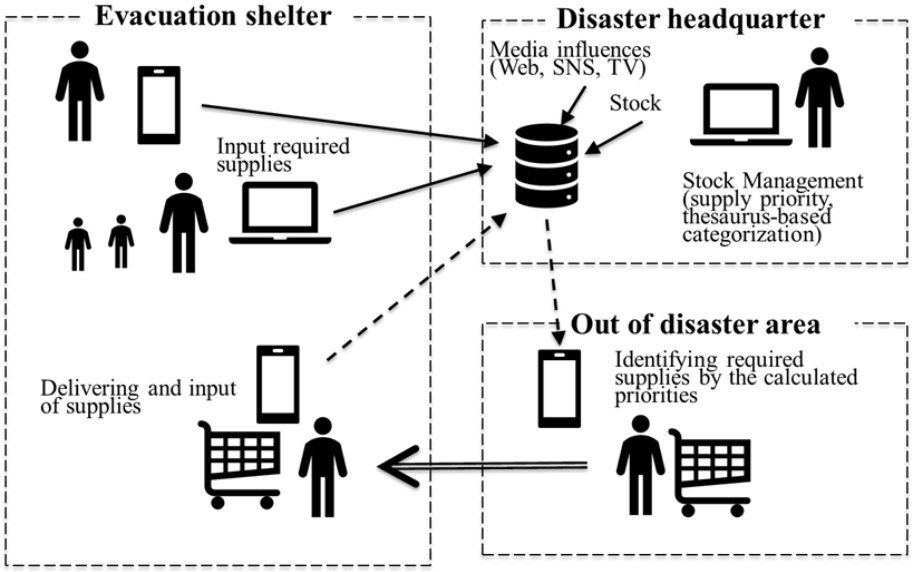


Figure 1: Example of Disaster Relief Supply Provision System

Based on these requirements, the assumed system must be capable of tracking the volume of stored relief supplies, the quantity of newly delivered items, and the transition of supplies required by disaster victims over time. A system configuration such as that illustrated in Figure 1 can support the integration of these elements and enable comprehensive supply management across different disaster response phases.

Furthermore, it is considered necessary to introduce an algorithm based on a Poisson distribution for each category of disaster relief supplies. By modeling the arrival and consumption patterns of supplies using this approach, the system can estimate appropriate delivery volumes and timing as shown in Formula 1 and Formula 2. Such a function can support decision making for identifying suitable relief supplies and can contribute to more efficient and balanced disaster support operations.

$$\boldsymbol{P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}} \tag{1}$$

$$\boldsymbol{\lambda = \lambda_{victims} - \lambda_{web} - \lambda_{stock}} \tag{2}$$

In the formulas, λ is the expected rate of required supply, X is a discrete random variable, k is the number of occurrences, e is Euler's number. Also, λ consists of the occurrence of required supply from victims, the occurrence from web or media, and the amount of supply stock in damaged area.

As future work, the proposed algorithm is planned to be applied to simulation studies using actual records from the 2024 Noto Peninsula Earthquake. In particular, simulations will be conducted to estimate required resource quantities, such as water supplies, and to compare them with delivered support levels. Through this process, the proposed method is expected to be systematically evaluated and refined.

6 Conclusions and Future Study

This paper reported the transition of required information during large scale disasters, with a particular focus on the Noto Peninsula Earthquake. By reviewing past major earthquakes in Japan and related studies, the analysis highlighted how disaster response has evolved together with changes in information and communication technologies. The findings confirmed that the needs of disaster victims change significantly over time, moving from immediate survival related information and supplies toward support for recovery and reconstruction.

The analysis of access logs from the official website of Nanao City demonstrated clear temporal changes in information demand. Immediately after the earthquake, access was concentrated on information related to lifelines, relief supplies, food, and water. Five months later, interest shifted toward donations, disaster waste management, and recovery support systems. These results provide concrete evidence that information demand follows a phased transition, which should be considered when designing disaster response systems.

Based on these observations, the paper discussed issues related to Disaster Relief Supply Provision System under modern communication environments. While the availability of communication networks and SNS has enabled rapid information sharing, it has also caused problems such as misinformation, excessive supply delivery, and mismatches between actual needs and provided supplies. The examples from recent disasters indicate that effective disaster response requires not only the delivery of information but also accurate management and coordination from victims, media and delivered supplies.

To address these challenges, this paper presented considerations for a Disaster Relief Supply Provision System that accounts for the transition of required supplies over time. Key requirements

include real-time understanding of victim needs with web or SNS, prioritization based on data analysis, and unified categorization of supply items. The introduction of probabilistic approaches, such as algorithms based on Poisson distributions, was discussed as a method for supporting decision making in supply allocation and timing.

As future study, further validation of the proposed system using previous disaster cases is required. In particular, quantitative evaluation using real operational data would help clarify the effectiveness of the proposed functions. Through the validation, the proposed method is expected to be systematically evaluated and refined. Through continued research in these directions, it is expected that more adaptive and efficient disaster response systems can be developed. Such systems have the potential to reduce confusion, improve supply distribution, and support faster and more effective recovery in future large-scale disasters.

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